Understanding the Transformer

A visual guide to attention mechanisms

Abstract

The Transformer is a neural network architecture that uses attention to represent relationships between tokens in a sequence.

Instead of processing words one at a time, it allows each position to gather information from other positions in parallel.

This short reading sample introduces the encoder-decoder structure and explains how scaled dot-product attention works.

1 Introduction

Language contains dependencies that can span many words: a pronoun may refer to a noun that appeared much earlier.

Attention creates a direct path between these positions, allowing the model to combine local details with wider context.

The original architecture contains an encoder that reads the input and a decoder that generates the output.

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2 Scaled Dot-Product Attention

Each token is mapped to a query, a key, and a value, conventionally written as Q, K, and V.

The dot product between a query and a key measures their compatibility.

Dividing the scores by the square root of the key dimension helps keep their magnitude manageable.

A softmax converts these scores into nonnegative weights that sum to one.

The weighted sum of values produces a context-sensitive representation for each query.

Figure 1 illustrates the flow from queries and keys to attention weights and the final output.

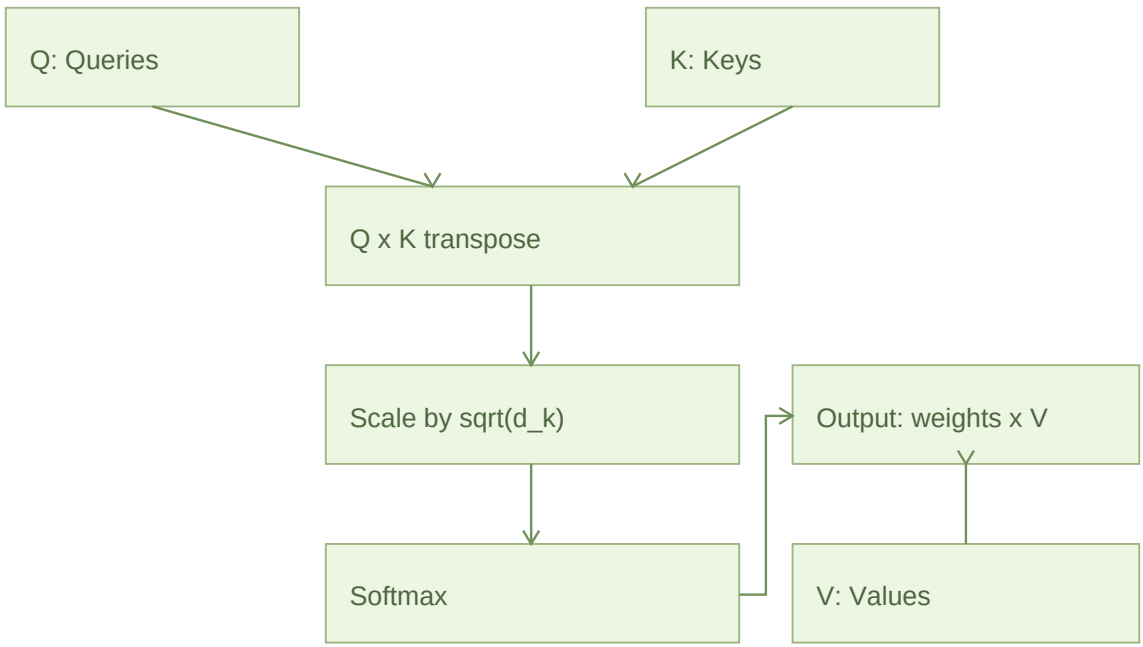

Figure 1. Attention score computation (simplified illustration).

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3 Multi-Head Attention

Multiple attention heads apply different learned projections to the same input.

This allows the model to represent several kinds of relationships at the same time.

The outputs of the heads are concatenated and passed through a final linear projection.

Positional information is also needed because attention alone does not specify the order of the tokens.

In practice, interpreting attention weights requires care: they describe one part of the computation rather than a complete explanation of the prediction.